

Multiple Choice

1. **Answer:** A

Options: A) 11.2 liters; B) 67.2 liters; C) 168 liters; D) 100.8 liters; E) 22.4 liters

Note: Correct option: A - 11.2 liters

2. **Answer:** B

Options: A) 0.159 pm; B) 79.5 pm; C) 210 pm; D) 132 pm; E) 0.53 pm

Note: Correct option: B - 79.5 pm

3. **Answer:** C

Options: A) 1000 H^+ ions; B) 900 H^+ ions; C) 843 H^+ ions; D) 950 H^+ ions; E) 615 H^+ ions

Note: Correct option : C – 843 H^+ ions

3. **Answer:** A

Options: A) It contains 20 protons and neutrons; B) All atoms of calcium have 20 protons and 40 neutrons; C) It contains 20 protons, neutrons, and electrons; D) All calcium atoms have 20 protons, neutrons, and electrons; E) All atoms of calcium have a mass of 40.078 u

Note: Correct option: A - It contains 20 protons and neutrons

4. **Answer:** C

Options: A) H-O; B) P-Cl; C) N-H; D) S-O; E) C-O

Note: Correct option: C - N-H

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Fe'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '250 liters'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' 5.75010^{14} Hz'.

Long Form Response

11. **Answer:** 28.4 g of Ca(OH)_2 , 25.1 g of H_3PO_4 . Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 0.15 J mol^{-1} . Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key